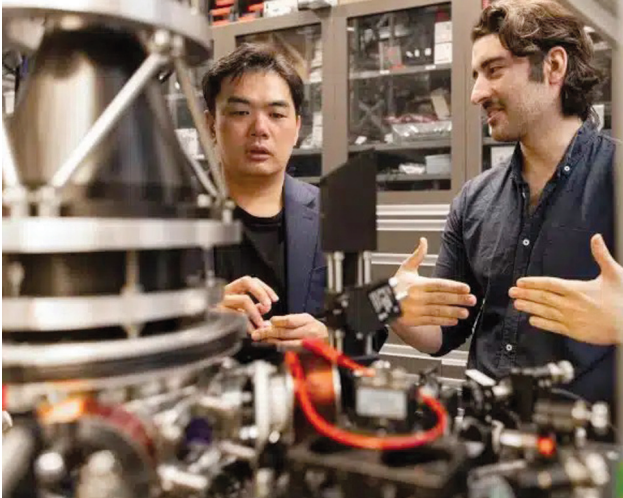


Error-correctable qubits and universal gates realised within an atom at room temperature

Physicists at the University of Sydney have achieved a breakthrough by demonstrating a universal set of logic



Dr Tingrei Tan (left) and his PhD student Vassili Matsos inspect the Paul trap used in this experiment in the Quantum Control Laboratory at the University of Sydney Nano Institute

gates for Gottesman-Kitaev-Preskill (GKP) qubits inside a single trapped ion. Using one ytterbium ion in a Paul trap, they encoded two logical qubits in its vibrational modes and executed single- and two-qubit operations. The advance reduces error-correction overhead, demonstrating that error-correctable qubits and universal gates can be realised within a single atom at room temperature.

ISRO succeeds in first integrated air drop test for Gaganyaan missions

ISRO (Bengaluru) has successfully conducted the first integrated air drop test (IADT-01) of the Gaganyaan crew module at Sriharikota, validating its parachute-based deceleration system for astronaut safety. A 4.8-tonne module was dropped from 3km using an IAF Chinook, deploying ten parachutes in sequence to reduce speed to 8m/s before splashdown. Recovered by the Navy, the test marks a key milestone toward Gaganyaan's uncrewed orbital mission planned for late 2025.



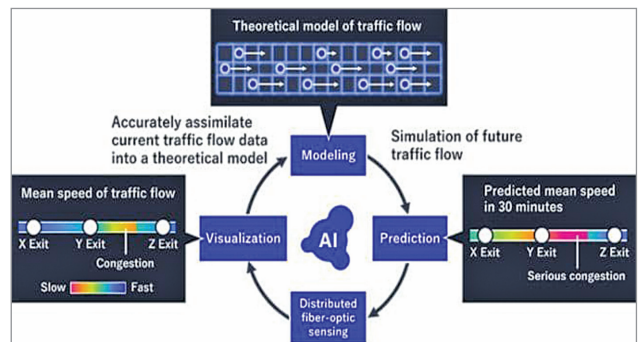
ISRO carries out first integrated air drop test for Gaganyaan missions (Credit: <https://www.wionews.com>)

2mm thick display for next-gen devices

Meta (California, United States) has unveiled a flat laser display just two millimetres thick, promising to revolutionise AR glasses and enhance visuals in smartphones, tablets, and TVs. Built around a centimetre-wide photonic chip paired with a tiny liquid-crystal-on-silicon panel, it shrinks bulky optics onto a fingernail-sized device while delivering high-resolution, vibrant images. Although challenges such as laser speckle and energy inefficiency remain, the innovation points toward ultra-thin, immersive, and energy-saving displays for next-generation devices.

Fibre-optic cable enables real-time traffic forecasting

NEC Corporation (Minato City, Tokyo, Japan) has developed an optical fibre sensing system that monitors road conditions and predicts sudden traffic congestion in real



Technology overview (Credit: <https://www.nec.com>)

time. By using existing roadside fibre-optic cables as distributed sensors, the system captures vehicle-induced changes across entire routes. Data is processed via NEC's AI, combining model parameter optimisation and data adaptation algorithms, enabling highly accurate traffic simulations. Trials have shown up to 80% fewer prediction errors, with deployment targeted for 2026.

Semantic communication framework paves way for 6G

Researchers at Seoul National University of Science and Technology (South Korea) have introduced ConcreteSC, a cutting-edge digital semantic communication framework designed to enhance the robustness, efficiency, and low-overhead capabilities of wireless systems. Unlike conventional vector quantisation, ConcreteSC uses temperature-controlled concrete distributions to generate bitstreams directly, enabling fully differentiable, noise-resilient training. Tested on ImageNet under fading channels, it outperformed VQ methods, scaling linearly with bit length. Its modular design promises real-world 6G deployment in smart factories, healthcare IoT, and AI-driven networks.